FarmGuard AI

Exploratory Data Analysis — Livestock Blood Biomarkers Dataset

This report explores the cleaned **Animals** dataset (750 records, 50 cattle across 5 genetic groups, sampled under 3 heat-stress periods defined by Temperature-Humidity Index range: T0 = low, T1 = moderate, T2 = high). The goal is to understand how blood biomarkers — especially cortisol, the primary physiological heat-stress marker — vary by breed and heat exposure, to inform FarmGuard AI's Livestock Heat Intelligence risk model.

1. Dataset Overview

Metric	Value
Total records	750
Unique animals	50
Genetic groups	5 (Local, HF50, HF62.5, HF75, HF87.5)
Heat-stress periods	3 (T0 = Low, T1 = Moderate, T2 = High THI)
Biomarker columns	9
Missing values	0
Duplicate rows	0
Design	Fully balanced — 50 records per genetic group × THI period

2. Summary Statistics

Descriptive statistics for all nine blood biomarkers across the full dataset.

Biomarker	Mean	Std	Min	25%	Median	75%	Max
Cortisol (µg/dL)	5.61	1.85	2.62	4.10	5.29	6.79	10.87
Glucose (mmol/L)	3.15	0.22	2.49	2.99	3.15	3.30	3.78
Total Protein (g/dL)	7.33	0.75	5.94	6.76	7.16	7.91	9.29
Uric Acid (mg/dL)	1.22	0.21	0.63	1.08	1.22	1.37	1.93
Cholesterol (mg/dL)	168.16	57.09	80.52	117.19	149.07	230.32	284.27
Calcium (mg/dL)	3.99	0.91	1.91	3.29	3.89	4.73	6.25
HDL (mg/dL)	106.66	29.77	47.85	71.38	116.80	132.12	156.12
AST (U/l)	26.42	3.36	17.93	23.67	26.21	28.85	35.19
ALT (U/l)	32.99	3.17	24.51	30.88	32.94	35.35	40.97

Outlier check (IQR method, 1.5×IQR fences): only 7 mild outliers total across 750×9 = 6,750 data points (2 in cortisol, 2 in glucose, 3 in uric acid) — the dataset is clean with no data-entry anomalies.

3. Distributions

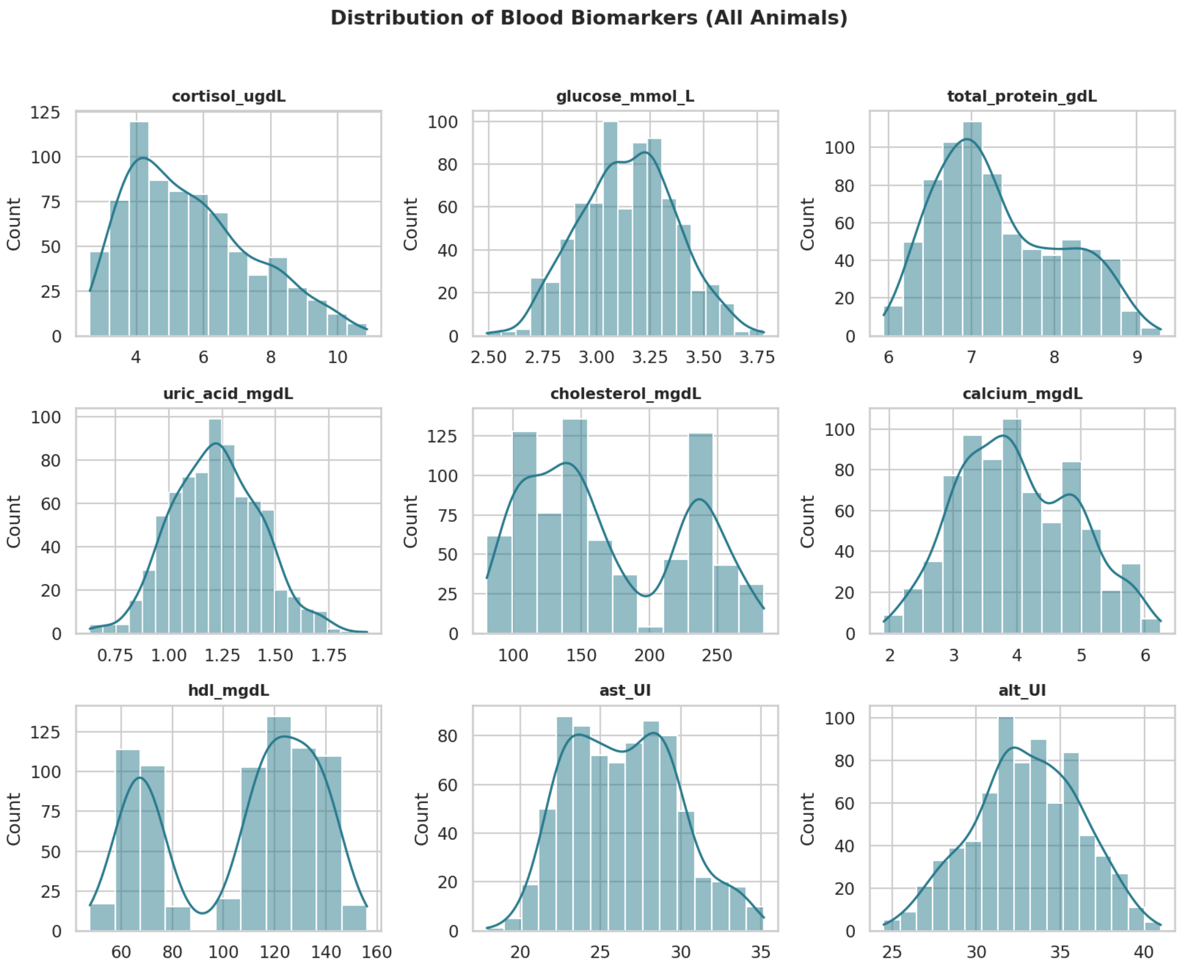

Figure 1. Distribution of all nine blood biomarkers across the full sample. Most biomarkers are approximately unimodal with no extreme skew; cholesterol and HDL show the widest spread, reflecting the range across genetic groups and heat-stress periods.

4. Heat Stress Effect on Cortisol

Cortisol is the primary physiological marker of heat stress. Its response to increasing THI is the clearest signal in the dataset.

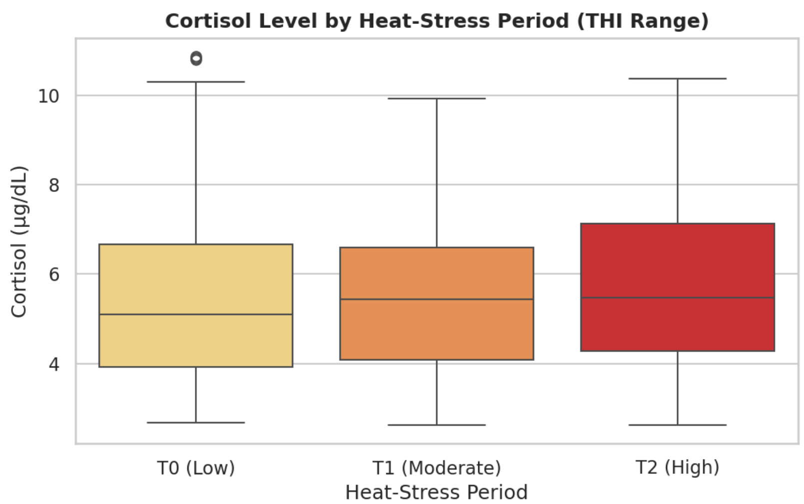

Figure 2. Cortisol rises consistently from T0 (low THI) to T2 (high THI): mean cortisol is 5.46 µg/dL at T0, 5.60 at T1, and 5.76 µg/dL at T2 — a steady increase confirming cortisol as a reliable heat-stress indicator.

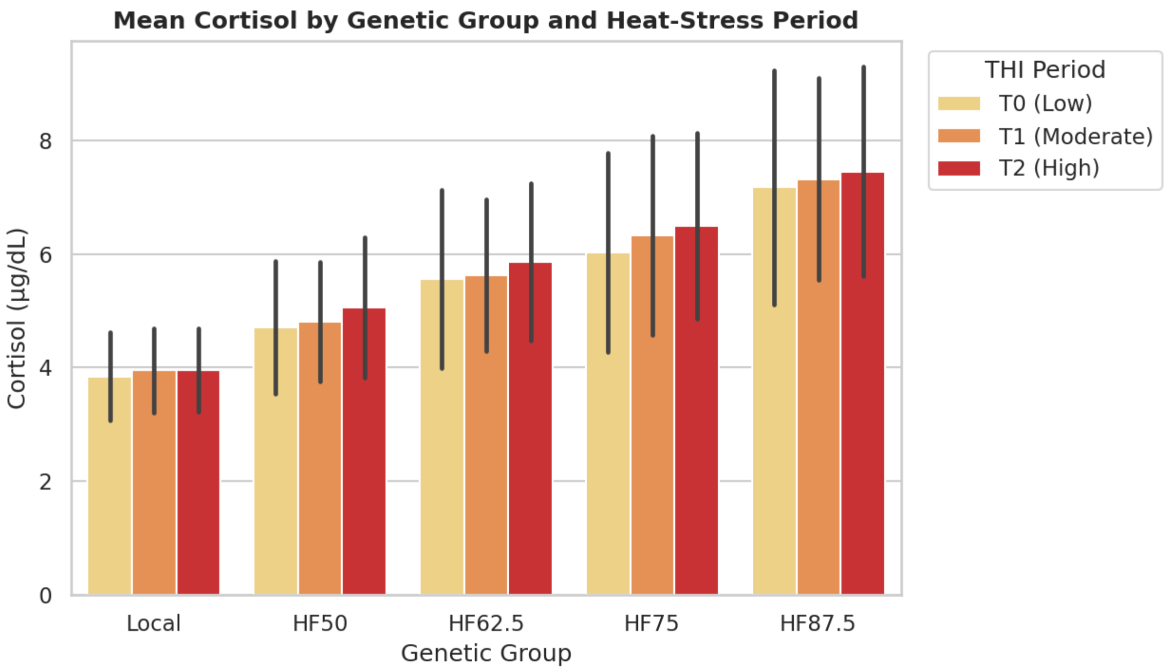

Figure 3. Mean cortisol by genetic group and heat-stress period. Baseline cortisol scales strongly with the proportion of Holstein-Friesian genetics (Local lowest, HF87.5 highest), and every group shows a further rise under heat stress (T0 → T2).

5. Breed Comparison — Heat Tolerance

Genetic Group	Cortisol T0	Cortisol T1	Cortisol T2	% Rise (T0→T2)
Local	3.85	3.95	3.96	2.8%
HF87.5	7.17	7.31	7.45	3.9%
HF62.5	5.56	5.62	5.86	5.4%
HF50	4.71	4.80	5.06	7.4%
HF75	6.03	6.32	6.50	7.8%

Ranked by relative cortisol increase from T0 to T2 (lowest = most heat-tolerant):

- **Local** breed shows the smallest relative rise (2.8%) — most heat-resilient physiologically, consistent with native adaptation to hot climates.
- **HF87.5** has the highest absolute cortisol at every period, but a moderate relative rise (3.9%).
- **HF50** and **HF75** show the largest relative jumps (7.4% and 7.8%), suggesting intermediate crossbred groups may be more physiologically reactive to heat.

Implication for FarmGuard AI: breed/genetic group should be a weighting factor in the Livestock Heat Intelligence risk score — the same THI level does not translate to equal stress across genetic groups.

6. Other Biomarkers Across Heat-Stress Periods

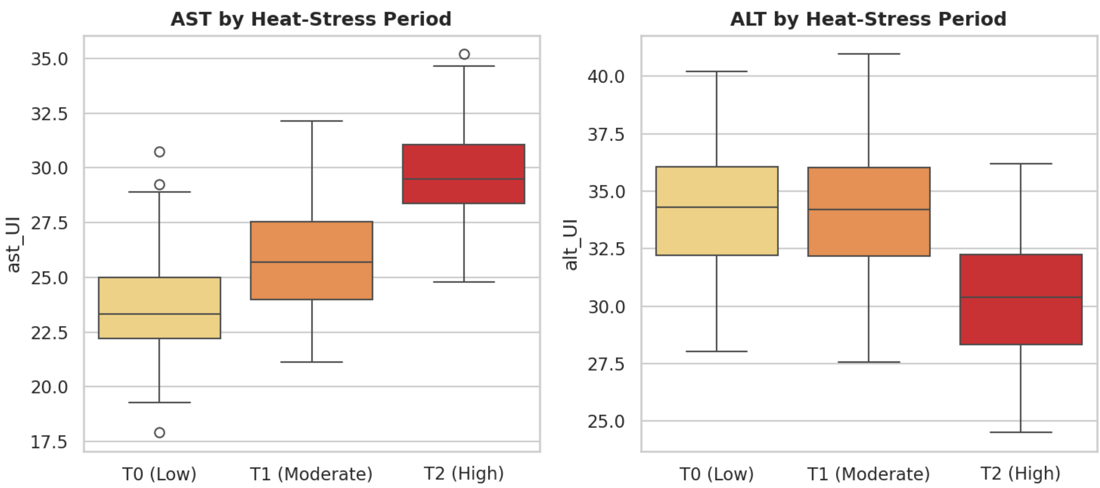

Figure 4. Liver enzymes AST and ALT both trend upward from T0 to T2, supporting cortisol as part of a broader stress response rather than an isolated signal.

Figure 5. Mean glucose and total protein by genetic group across heat-stress periods. Trends are less uniform than cortisol, consistent with glucose showing a weak negative correlation with cortisol (see Section 7).

7. Correlation Between Biomarkers

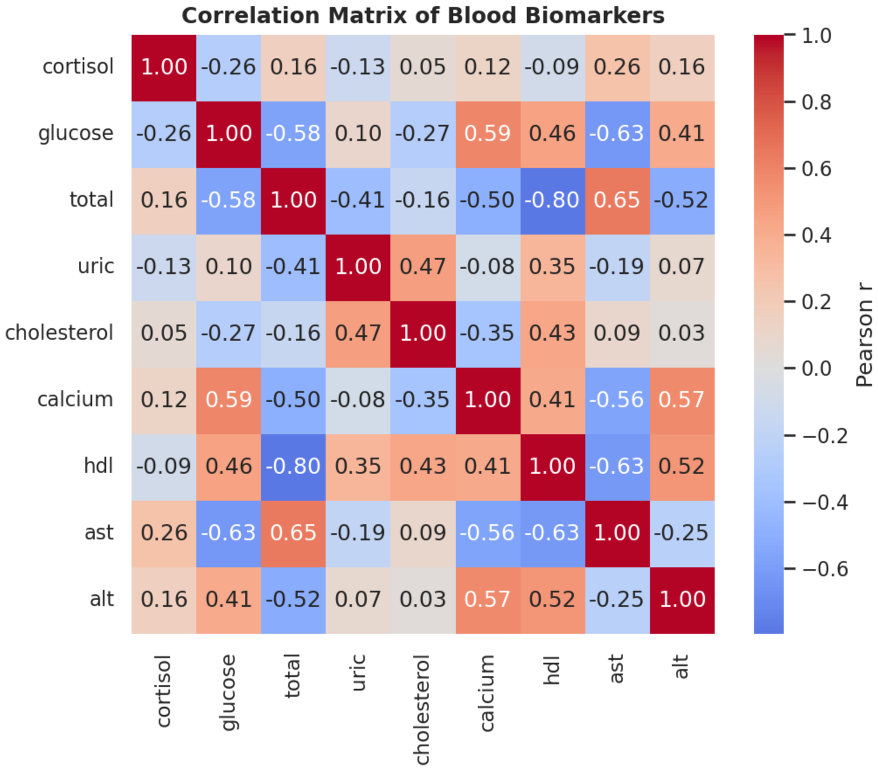

Figure 6. Pearson correlation matrix across all nine biomarkers.

Strongest relationships with cortisol (the key stress marker):

• AST (+0.26) and ALT (+0.16) — positive, consistent with heat-stress-linked liver enzyme activity.

• Glucose (–0.26) — negative; higher cortisol periods associate with slightly lower glucose, possibly reflecting altered feed intake under heat stress.

• Uric acid (–0.13) and HDL (–0.09) — weak negative associations.

• Calcium, total protein, and cholesterol show weak/near-zero correlation with cortisol individually.

8. Key Takeaways

• The dataset is clean and balanced — no missing values, no duplicates, only 7 mild outliers out of 6,750 values.

• Cortisol increases consistently with THI across all genetic groups, validating it as the core heat-stress signal for the risk model.

• Genetic group matters: Local cattle are the most heat-tolerant (smallest relative cortisol rise), while HF87.5 has the highest absolute stress load at every THI level.

• AST and ALT move with cortisol, reinforcing a coherent multi-marker stress signature useful for validating any single-metric risk score.

• Recommendation: FarmGuard AI's Livestock Heat Intelligence module should weight risk recommendations by genetic composition, not THI alone.